

# **Mathematics Bootcamp**

## **General Equilibrium and the Social Planner**

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# Today

- **The household:** an infinite path, boundary conditions, shadow values
- **The Euler equation**, and the firm problem
- **Competitive equilibrium:** definition, characterization, and market clearing
- **Planner:** efficiency, supporting prices, and existence

# From a Point to a Path

Day 4 studied a scalar choice

$$x \in D \subseteq \mathbb{R}.$$

Today the choice is an infinite sequence:

$$z = (x_0, x_1, x_2, \dots).$$

In the model that follows, a candidate path satisfies one Euler equation at every date and one transversality condition at infinity.

# Dated Commodities

A commodity is identified not only by its physical features, but also by its date and state. In this deterministic model, date is the key label.

## Different Date, Different Good

Consumption at  $t$  and  $t + 1$  are distinct commodities. Saving exchanges the current commodity for a future one; the interest rate will determine their relative price.

# Two Market Structures

## Arrow–Debreu

- Markets open only at date 0.
- Every dated good is traded there, at prices  $\{p_t\}$ .
- A single lifetime budget constraint.
- Perfect commitment: contracts signed at 0 are honored.

## Sequential Markets

- Markets reopen at every date.
- Spot goods and one-period assets, at prices  $\{w_t, r_t\}$ .
- One budget constraint per date.
- Debt must be bounded by a no-Ponzi condition.

Both assume perfect information and perfect competition, and both deliver the same consumption allocations. We work with sequential markets.

# The Environment

## Dynamic Production Economy

The environment is

$$E = (\mathbb{N}_0, \beta, u, f, (A_t)_{t=0}^{\infty}, k_0) .$$

These primitives are fixed before prices and allocations are determined. They define the **neoclassical growth model**, our canonical example for the rest of the day.

### Primitives

- Dates  $t = 0, 1, \dots$
- Preferences:  $\beta$  and  $u$
- Technology:  $f$  and  $A_t$

### Initial Resources

- The household owns  $a_0 = k_0 > 0$ .
- Labor supply is one unit each date.
- Capital depreciates at rate  $\delta \in (0, 1]$ .

# Lifetime Utility

## Lifetime Utility

For  $c = (c_0, c_1, c_2, \dots)$ ,

$$U(c) = \sum_{t=0}^{\infty} \beta^t u(c_t), \quad 0 < \beta < 1.$$

The discount factor  $\beta$  gives the relative weight on later utility. We assume that this sum is well-defined and finite on admissible paths.

## Monotonicity

$$u'(c_t) > 0$$

More consumption is preferred.

## Strict concavity

$$u''(c_t) < 0$$

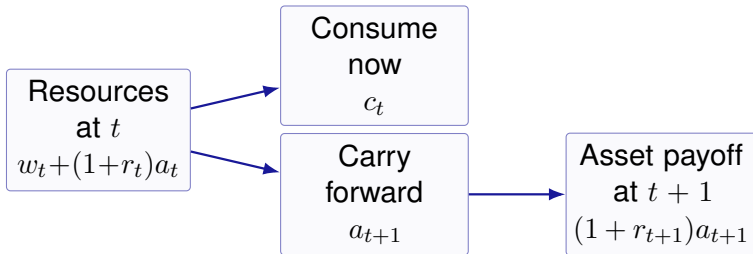
Marginal utility falls with consumption.

# Saving Transfers Resources Across Dates

At date  $t$ , the household owns assets  $a_t$ , supplies one unit of labor, and receives wage and capital income.

## Period Budget Constraint

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t.$$





# The Household Problem

Given  $\{w_t, r_t\}_{t=0}^{\infty}$  and  $a_0$

$$\max_{\{c_t, a_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to, for every  $t$ ,

$$c_t + a_{t+1} = w_t + (1 + r_t)a_t, \quad c_t \geq 0.$$

## State and Control

The **state**  $a_t$  is predetermined at date  $t$ ; the **control**  $c_t$  is chosen there. The budget constraint is the law of motion  $a_{t+1} = w_t + (1 + r_t)a_t - c_t$ .

# Boundary Conditions

## An Infinite Path Needs Both Ends

$$a_0 \text{ given, } \quad \lim_{t \rightarrow \infty} \frac{a_{t+1}}{\prod_{i=1}^t (1 + r_i)} = 0.$$

## Where the Terminal Condition Comes From

**No Ponzi**, imposed on the problem:  $\lim a_{t+1} / \prod (1 + r_i) \geq 0$ . Without it the household rolls debt over forever and no maximum exists.

**Transversality**, implied by optimality:  $\lim a_{t+1} / \prod (1 + r_i) \leq 0$ . Leaving value unspent is never optimal.

# Shadow Values

There is one budget constraint at each date, so the Lagrangean carries one multiplier at each date:

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t \left[ \underbrace{w_t + (1 + r_t)a_t}_{\text{resources at } t} - c_t - a_{t+1} \right].$$

## What $\lambda_t$ Measures

$\lambda_t$  is the shadow value of wealth at date  $t$ : the gain in lifetime utility from one more unit of the good available at  $t$ . Differentiating in  $c_t$  gives

$$\beta^t u'(c_t) = \lambda_t.$$

## The State Across Two Dates

The state  $a_{t+1}$  is chosen at  $t$  and carried into  $t + 1$ , so it appears in two constraints:

$$\underbrace{c_t + a_{t+1} = w_t + (1 + r_t)a_t}_{\text{date } t, \text{ multiplier } \lambda_t}, \quad \underbrace{c_{t+1} + a_{t+2} = w_{t+1} + (1 + r_{t+1})a_{t+1}}_{\text{date } t+1, \text{ multiplier } \lambda_{t+1}}.$$

Setting  $\partial L / \partial a_{t+1} = 0$  therefore couples the two shadow values:

$$\lambda_t = (1 + r_{t+1}) \lambda_{t+1}.$$

### Why This Is the Euler Equation

Saving one unit costs  $\lambda_t$  today and returns  $(1 + r_{t+1})\lambda_{t+1}$  tomorrow. Substituting  $\lambda_t = \beta^t u'(c_t)$  eliminates the multipliers.

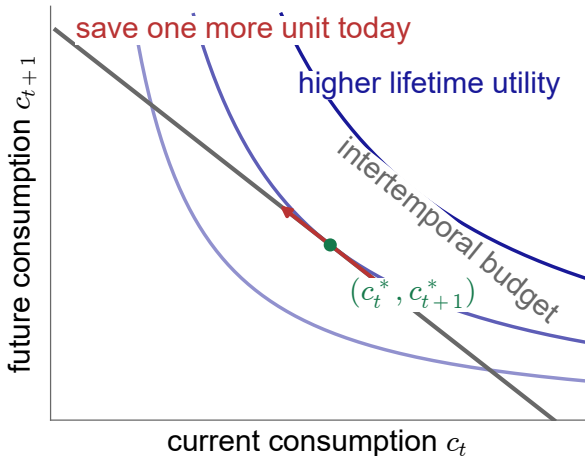
# Tangency Across Dates

The household treats consumption at  $t$  and  $t + 1$  as two different goods. At an interior optimum,

$$\frac{\beta u'(c_{t+1})}{u'(c_t)} = \frac{1}{1 + r_{t+1}}.$$

## Same Mathematics as Day 4

The MRS equals the relative price. The only new feature is that the goods carry date labels.



# Saving One More Unit: Marginal Cost Equals Marginal Benefit

Combine the two first-order conditions:

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).$$

## Cost of Saving One More Unit

One unit is forgone today, reducing lifetime utility by

$$\beta^t u'(c_t).$$

## Benefit One Date Later

Tomorrow it yields  $1 + r_{t+1}$  units. Lifetime utility rises by

$$\beta^{t+1}(1 + r_{t+1})u'(c_{t+1}).$$

At an interior optimum, these two marginal effects are equal.

# The Euler Equation and the Tail

For the interior candidate path in this model, impose the model-specific transversality condition

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) a_{t+1} = 0.$$

Same condition as on *Boundary Conditions*:  $\lambda_t = \beta^t u'(c_t)$  and  $\lambda_t = \lambda_0 / \prod_{i=1}^t (1 + r_i)$ , so the two limits differ only by the constant  $\lambda_0$ .

## Euler Equation

Compares the marginal cost and benefit of moving one unit between two adjacent dates.

## Transversality Condition

Rules out tails that retain resources with positive shadow value.

Neither condition proves existence. Feasibility, concavity, and a separate existence argument remain necessary.

# Technology

In every date  $t$ , the firm rents capital  $k_t$ , hires labor  $n_t$ , and produces the single final good:

$$y_t = A_t f(k_t, n_t) = A_t k_t^\alpha n_t^{1-\alpha}, \quad 0 < \alpha < 1.$$

## Technology

$A_t$  is total factor productivity. It scales output at every input pair;  $f$  remains the standard Cobb–Douglas function.

## Constant Returns

Under constant returns, any finite profit-maximizing plan earns zero profit. Positive profit would scale without bound.



# The Firm Problem

**Given  $w_t$  and  $r_t$ , at each date separately**

$$\max_{k_t, n_t \geq 0} \{k_t^\alpha (A_t n_t)^{1-\alpha} - w_t n_t - (r_t + \delta)k_t\}.$$

**Static, and Prices Equal Marginal Products**

There is no intertemporal link: the firm rents, produces and sells within date  $t$ . At an interior optimum,

$$w_t = A_t f_2(k_t, n_t), \quad r_t + \delta = A_t f_1(k_t, n_t).$$

# Factor Prices and the Return to Capital

For Cobb–Douglas production,

$$w_t = (1 - \alpha)A_t k_t^\alpha n_t^{1-\alpha}, \quad r_t + \delta = \alpha A_t k_t^{\alpha-1} n_t^{1-\alpha}.$$

A fraction  $\delta$  of the machine wears out in production, so the rental must cover depreciation on top of the net return. Save one unit today, receive  $1 + r_t$  tomorrow. Constant returns and the first-order conditions imply

$$A_t f(k_t, n_t) = w_t n_t + (r_t + \delta)k_t,$$

so competitive profits are zero.

# Feasible Allocations

## Allocation and Feasibility

A physical allocation is a sequence

$$z = (c_t, k_{t+1}, n_t)_{t=0}^{\infty}.$$

It is **feasible** if, for every  $t = 0, 1, \dots$ ,

$$c_t \geq 0, \quad k_{t+1} \geq 0, \quad 0 \leq n_t \leq 1,$$

and

$$c_t + k_{t+1} \leq A_t k_t^{\alpha} n_t^{1-\alpha} + (1 - \delta)k_t.$$

Feasibility is physical and contains no prices. Monotonicity makes the resource inequality bind at an optimum.

# Correspondences

## Household Correspondence

At prices  $(w, r)$ , write

$$H(w, r) = \{\text{household-optimal paths}\}.$$

It may contain more than one path  $(c, a)$ .

## Firm Correspondence

At prices  $(w_t, r_t)$ , write

$$F_t(w_t, r_t) = \{\text{optimal input pairs at } t\}.$$

It may contain more than one pair  $(k, n)$ .

Under constant returns,

positive profit  $\implies F_t(w_t, r_t) = \emptyset$ ,      zero profit  $\implies F_t(w_t, r_t)$  may be set-valued.

Equilibrium selects mutually compatible elements of  $H$  and  $F_t$ .

# Equilibrium: The General Definition

## The Shape of Any Equilibrium Definition

Given the primitives and the initial state, an equilibrium is a list of *quantities* and *prices* such that

1. households **optimize**, taking prices as given;
2. firms **optimize**, taking prices as given;
3. markets **clear**.

A definition *names* the objects and *states* the conditions. It uses no derivatives, and it does not claim that such a list exists.

It is a *test*, not an adjustment story: it says which lists qualify, not how an economy would find one.

# Equilibrium: The Actual Definition

*The general definition, filled in for the growth model.*

## Competitive Equilibrium

Given  $a_0$ , a competitive equilibrium consists of household allocations  $\{c_t, a_{t+1}\}_{t=0}^{\infty}$ , firm allocations  $\{k_t, n_t\}_{t=0}^{\infty}$  and prices  $\{w_t, r_t\}_{t=0}^{\infty}$  such that

1. given  $\{w_t, r_t\}$ , the household allocation solves the **household problem**;
2. given  $\{w_t, r_t\}$ , the firm allocation solves the **firm problem** at every date;
3. **markets clear** at every date.

The firm pays  $r_t + \delta$  per unit of capital, so the wage and the interest rate describe the whole price system.

# Market Clearing

## Three Markets, Every Date

$$\underbrace{a_t = k_t}_{\text{capital}}, \quad \underbrace{n_t = 1}_{\text{labor}}, \quad \underbrace{c_t + k_{t+1} = k_t^\alpha (A_t n_t)^{1-\alpha} + (1 - \delta)k_t}_{\text{goods}}.$$

## Walras' Law

If both problems are solved and two of the three markets clear, the third clears automatically. Goods-market clearing is implied by the other two.

# Equilibrium: Characterization

## Definition versus Characterization

**The definition** *states* conditions: solve, solve, clear.

**The characterization** *derives* what the conditions imply: the first-order conditions of each problem, joined to the market-clearing equations.

This is where derivatives enter, and nowhere earlier.

Existence and uniqueness are separate questions; a characterization settles neither.



# The Equilibrium System

## Every Equilibrium Path Satisfies, at Every Date

|                  |                                                    |                                                            |
|------------------|----------------------------------------------------|------------------------------------------------------------|
| <i>household</i> | $u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}),$         | $c_t + a_{t+1} = w_t + (1 + r_t)a_t,$                      |
| <i>firm</i>      | $w_t = (1 - \alpha)A_t k_t^\alpha n_t^{1-\alpha},$ | $r_t + \delta = \alpha A_t k_t^{\alpha-1} n_t^{1-\alpha},$ |
| <i>markets</i>   | $a_t = k_t,$                                       | $n_t = 1.$                                                 |

Six equations each date for the six unknown sequences  $(c_t, a_{t+1}, k_t, n_t, w_t, r_t)$ . The initial condition  $a_0 = k_0$  and the transversality condition select the path; goods-market clearing then follows by Walras' law.

# Prices as Functions of Capital

To reduce the system, set  $A_t = 1$  and substitute  $n_t = 1$ .

## General Technology

$$\begin{aligned}w_t &= f_n(k_t, 1), \\r_t &= f_k(k_t, 1) - \delta.\end{aligned}$$

## Cobb–Douglas Technology

$$\begin{aligned}w_t &= (1 - \alpha)k_t^\alpha, \\r_t &= \alpha k_t^{\alpha-1} - \delta.\end{aligned}$$

Firm optimality and market clearing remove prices as independent unknowns.

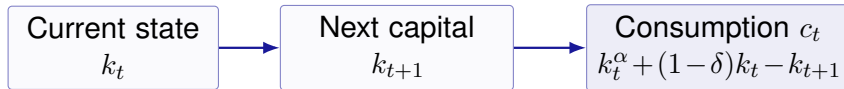
# Consumption from Capital

With  $n_t = 1$  and  $A_t = 1$ , the resource constraint is

$$c_t + k_{t+1} = k_t^\alpha + (1 - \delta)k_t.$$

Hence,

$$c_t = k_t^\alpha + (1 - \delta)k_t - k_{t+1}.$$



Once the capital path is known, the allocation and prices can be recovered.

# Three Substitutions Produce the Capital Recurrence

(1) Firm optimality replaces the return:

$$r_{t+1} = \alpha k_{t+1}^{\alpha-1} - \delta.$$

(2) Feasibility replaces consumption at both dates:

$$c_j = k_j^\alpha + (1 - \delta)k_j - k_{j+1}, \quad j = t, t + 1.$$

(3) Insert both expressions into the household Euler equation:

$$\begin{aligned} u'(k_t^\alpha + (1 - \delta)k_t - k_{t+1}) &= \beta [1 + \alpha k_{t+1}^{\alpha-1} - \delta] \\ &\quad \times u'(k_{t+1}^\alpha + (1 - \delta)k_{t+1} - k_{t+2}). \end{aligned}$$

The result is a second-order difference equation in  $(k_t, k_{t+1}, k_{t+2})$ .

# Necessity and Sufficiency

## Necessary Direction

interior equilibrium  $\implies$  capital recurrence.

## Converse Direction

capital recurrence + feasibility + initial condition  
+ transversality + sufficient optimality conditions  $\implies$  equilibrium.

Neither implication proves that a solution exists or that it is unique.

# Pareto Efficiency

## Efficiency

A feasible allocation  $z^* = (c_t^*, k_{t+1}^*, n_t^*)_{t=0}^{\infty}$  is efficient if there is no feasible  $\tilde{z}$  such that  $U(\tilde{c}) > U(c^*)$ .

Efficiency uses only preferences and physical feasibility: no prices and no ownership rule. With one household, the planner selects the efficient path by maximizing the household's lifetime utility over feasible allocations.

# The Planner Problem

With  $A_t = 1$ , the planner solves

$$\max_{(c_t, k_{t+1}, n_t)_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$\begin{aligned} c_t + k_{t+1} &= k_t^{\alpha} n_t^{1-\alpha} + (1 - \delta)k_t, & t = 0, 1, \dots, \\ c_t \geq 0, \quad k_{t+1} \geq 0, \quad 0 \leq n_t \leq 1, \quad k_0 > 0 & \text{ given.} \end{aligned}$$

Since leisure does not enter utility and production increases in labor,  $n_t = 1$ . At an interior optimum,

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) k_{t+1} = 0.$$

# The Planner Euler Equation

For the reduced problem, define

$$L = \sum_{t=0}^{\infty} \beta^t u(c_t) + \sum_{t=0}^{\infty} \lambda_t [k_t^\alpha + (1 - \delta)k_t - c_t - k_{t+1}].$$

The first-order conditions give

$$\lambda_t = \beta^t u'(c_t), \quad t = 0, 1, \dots,$$

$$\lambda_t = \lambda_{t+1} [\alpha k_{t+1}^{\alpha-1} + 1 - \delta], \quad t = 0, 1, \dots$$

Therefore,

$$\boxed{u'(c_t) = \beta [\alpha k_{t+1}^{\alpha-1} + 1 - \delta] u'(c_{t+1}).}$$



# Comparing the Two Euler Equations

## Competitive Equilibrium

Household Euler equation:

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}).$$

Firm return and clearing:

$$1 + r_{t+1} = 1 + \alpha k_{t+1}^{\alpha-1} - \delta.$$

=

## Social Planner

Planner return:

$$1 + \alpha k_{t+1}^{\alpha-1} - \delta.$$

Planner Euler equation:

$$u'(c_t) = \beta [1 + \alpha k_{t+1}^{\alpha-1} - \delta] u'(c_{t+1}).$$

The equality is economically powerful only because concavity and transversality make these equations sufficient.

# The Market and Planner Choose the Same Allocation

Assume lifetime utility is well-defined, the relevant optima are interior, the problems are concave, and both transversality conditions hold.

## Equilibrium $\Rightarrow$ Planner

Every equilibrium allocation is a planner optimum:

equilibrium  $\Rightarrow$  planner optimum.

Markets attain an efficient allocation.

## Planner $\Rightarrow$ Equilibrium

Every interior planner optimum can be supported by prices:

planner optimum  $\Rightarrow$  equilibrium.

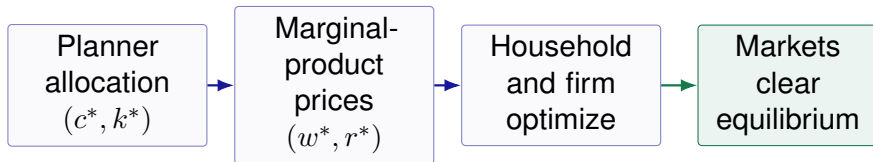
Marginal products construct the prices.

For an infinite horizon, these assumptions are part of the theorem; the Euler equations alone are not sufficient.

# Supporting Prices

Start from an interior planner solution  $(c_t^*, k_{t+1}^*)_{t=0}^\infty$  and define

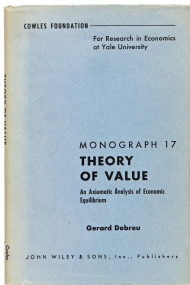
$$w_t^* = (1 - \alpha)(k_t^*)^\alpha, \quad r_t^* = \alpha(k_t^*)^{\alpha-1} - \delta.$$



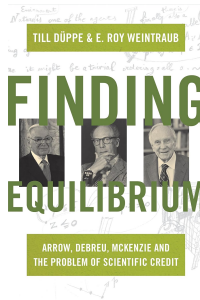
- (1) Concavity makes the firm's marginal-product conditions sufficient.
- (2) The planner Euler equation and transversality verify household optimality.
- (3) Set  $a_t^* = k_t^*$  and  $n_t^* = 1$ ; planner feasibility is goods clearing.

# Existence of Equilibrium

The definition says what an equilibrium *is*, not that one exists. Existence is a separate theorem: Arrow and Debreu (1954) and, independently, McKenzie (1954), by fixed-point arguments.



Debreu, *Theory of Value* (1959). Cowles Monograph 17.



Düppe and Weintraub, *Finding Equilibrium* (2014).